Compare seed treatment types

Choose components for the crop and target problem. Compare composition and labeled uses before comparing brands. This reference is educational; it does not rank products or replace crop-specific Extension advice.

Treatment type	Primary purpose	What to compare	Limits / considerations
Fungicide	Protect seed and seedlings from specified seedborne or soilborne pathogens.	Compare labeled pathogens, ingredients, and FRAC groups.	Protection varies by pathogen; no single ingredient covers every seedling disease.
Insecticide	Manage specified insects during crop establishment.	Use pest history and crop-specific IPM guidance; compare IRAC groups.	Benefits depend on pest pressure. Early protection does not imply season-long control.
Nematicide	Reduce early root exposure to specified plant-parasitic nematodes.	Use nematode testing and labeled species claims.	Combine with resistant varieties and rotation; seed treatment alone may be inadequate.
Biological biopesticide	Use a microorganism or qualifying natural substance to manage a labeled pest.	Check the exact strain or substance and target pest.	The biological category can overlap with the functions above; follow its label.
Inoculant	Support legume nodulation and biological nitrogen fixation.	Check crop compatibility and field history.	An inoculant is not automatically a pesticide. Verify compatibility with other seed-applied products.

Biological describes an origin or technology and can overlap with fungicide, insecticide, or nematicide functions. Plant-incorporated protectants (PIPs) are produced within the plant; they are not seed coatings. The EPA site census can also include fumigants and other technologies; inclusion is not a recommendation to apply them as a seed coating.

State approval check: Federal EPA registration does not establish state registration. Check the exact product and EPA number for the treatment and planting locations. Provider availability reports are not official approval; missing data means not verified.

Product availability, application methods, crop uses, and efficacy must be verified separately. Inoculants are included for comparison; the EPA pesticide export is not a complete inventory of inoculant products.

Sources: CPN: Factors to Consider Before Using a Soybean Seed Treatment (updated March 19, 2026)
EPA: What are Biopesticides?

Read the resistance groups

FRAC and IRAC are separate systems. Codes describe resistance-management groups, not safety, efficacy, or approved uses. Unmapped ingredients remain unclassified; this does not mean they have no group or no resistance risk. Biological strain and plant-incorporated protectant assignments require separate review.

FRAC fungicide groups - selected examples

Group	Class / group	Example active ingredients
1	MBC	Thiabendazole; thiophanate-methyl
3	DMI	Tebuconazole; ipconazole; difenoconazole
4	Phenylamide	Metalaxyl; metalaxyl-M (mefenoxam)
7	SDHI	Sedaxane; fluopyram; fluxapyroxad
11	QoI	Azoxystrobin; pyraclostrobin; trifloxystrobin
12	Phenylpyrrole	Fludioxonil
22	Benzamide	Ethaboxam
49	OSBPI	Oxathiapiprolin
M03	Multiple sites	Mancozeb; thiram
U17	Unknown mode of action	Picarbutrazox

IRAC insecticide groups - selected examples

Group	Class / group	Example active ingredients
1B	Organophosphate	Acephate
3A	Pyrethroid	Bifenthrin; tefluthrin
4A	Neonicotinoid	Clothianidin; imidacloprid; thiamethoxam
6	Avermectin	Abamectin
28	Diamide	Chlorantraniliprole; cyantraniliprole
30	GABA allosteric modulator	Broflanilide

Changing brands does not necessarily change the mode of action. Count only components effective against the target pest. Mixtures do not replace a resistance-management plan. Follow label restrictions and crop-specific Extension guidance across seed, soil, and foliar uses.

Examples are illustrative, not a complete code list or a crop-use recommendation. Ingredient group assignments do not establish a labeled seed-treatment use. See the database for the expanded reviewed reference.

Sources: FRAC Code List 2026
IRAC Mode of Action Classification v11.5 (February 2026)

Signal words, warnings & BMPs

Signal words describe acute hazard of the formulated pesticide as sold. They do not rank efficacy or summarize chronic or environmental risks. Do not infer POISON or a skull-and-crossbones from DANGER alone. EPA workbook values are not a review of the current label; an SDS signal word is not a substitute for the EPA label signal word.

EPA signal word / field	Meaning	Read with this caution
DANGER	Highest acute toxicity category I in at least one assessed endpoint.	POISON / skull-and-crossbones depends on the basis for category I; consult the label.
WARNING	Highest acute toxicity category II.	Read the hazard statements, PPE, and first-aid directions.
CAUTION	Category III; may also be used voluntarily for category IV.	Does not mean harmless or suitable for every use.
None reported by EPA	The export explicitly contains "None."	Category IV does not require a signal word, but do not infer a product's category from this field alone.
Not supplied	The export field is blank, absent, or unrecognized.	Unknown is not equivalent to no hazard. Check that registration's label.

The database keeps explicit "None" separate from missing data. Restricted-use flags come from the same registration's EPA row. Blank distributor warnings are never inherited from the base product. Product-specific precautions still require current-label review.

Best management practices

1. Start with a pest-management need. Use crop and field history, scouting or diagnostic tests, and local Extension guidance to choose components for the target problem.

2. Check the current label and seed tag. Verify crop, target pest, rates, application method, PPE, worker protections, planting restrictions, and first aid. Follow applicable restricted-use requirements.

3. Control abrasion and dust. Handle bags gently; maintain planting equipment. Use seed-flow lubricants and equipment settings permitted by the label and manufacturer.

4. Protect pollinators and water. Consider flowering areas, bee activity, wind, and nearby water. Avoid loading or cleaning where dust can reach bees; follow label precautions.

5. Clean up and keep seed segregated. Promptly address spilled seed and dust as directed. Keep treated seed out of food, feed, and oil channels. Follow label storage and disposal directions.

6. Plan for resistance and compatibility. Review FRAC/IRAC groups across the crop program. Check compatibility and storage instructions for each component, including inoculants. Keep product and treatment records.

Use applicable U.S. labels and local requirements. The Canadian source supports general stewardship practices; its legal requirements are not presented as U.S. rules.

Sources: 40 CFR 156.64: EPA signal-word requirements
EPA: Pesticide labels
EPA: Pollinator protection BMP resources
Health Canada: Treated-seed pollinator BMPs (general practices; Canadian rules are not U.S. requirements)